

Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing

Week 1 – Project Selection and Dataset Discovery

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Submitted By:- G9 - The Medics.

I. WORK DONE THIS WEEK:-

This week we majorly worked on Project Selection. Our Group decided to go with Brain MRI Tumor Segmentation, as Medical Image Processing is one of the oldest and classical examples of Image Processing. After looking at 20 topics, we decided to go with this topic because the technology in the Medical Domain has expanded to that level that a need for arranged systematic image processing is required.

Cancer is currently one of the most deadly diseases due to the lack of an actual cure, using Chemotherapy as a form to just counter it. Cancer basically is a condition when cells develop a mutation and start dividing uncontrollably and dying after the division, forming a lump of dead cells. There are also Non Cancerous Tumors, but Cancer is a more serious problem. In a brain tumor, the mass of cells grows relatively slower and stays in one place, whereas cancer spreads fast and attempts to destroy all cells in its near vicinity, spreading to other parts of the body too if possible.

Through this principle, we felt that looking into samples from the domain of Brain Tumor, it would help any patients while we work on the contour differences among the images. We understood that you cannot directly work on MRI scans, those have to be converted to Greyscale images like we learnt in class. We would have to apply filters to remove any noisy elements from the image, applying various noise models to see which applies the best for the given image. We then have to apply Segmentation to differentiate between the Tumor, Skull, Brain Matter.

We could later apply filtering again to differentiate between the healthy tissues and the sick masses growing throughout the brain. Working as a Segmentation among the Brain Cells

using Gaussian Segmentation, this could be done according to the intensity of those cells, Healthy Cells tend to have a lighter intensity, and dead cells tend to be darker, according to what we have learnt yet.

We might be able to use many other methods that we learn in the future of this course, but this is how we were able to relate our current learning, we also discovered the following dataset (attached below) and are looking into which of the other ones we should focus on more, the dataset contains direct fMRI data from Tumor Patients.

Dataset Taken from:-

<https://openneuro.org/datasets/ds005003/versions/2.0.0/>

Example image from the First Dataset:-

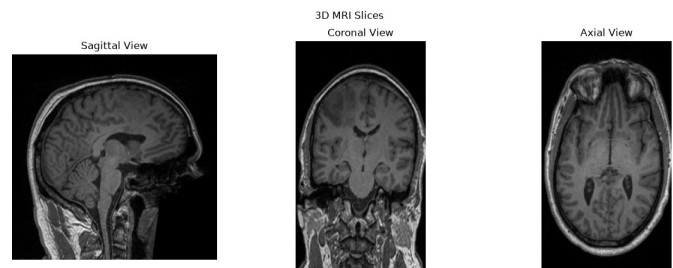


Fig. 1. Dataset Snapshots

II. WORK TO BE DONE NEXT WEEK:-

Next Week, We plan to start working on the coding aspect related to the approved Dataset, working on basic conversion of the images to complete Greyscale so that we can further work on it as we have learnt in class, trying to remove any noise associated with the images, and progress as further as we can according to the time we have got.

We also plan to look more deeply into how other scientists and researchers have worked on the given field before us, This Submission we are facing a lack of time to actually attempt a proper understanding of the actual concepts, we are rather relying on concepts learnt from prior courses in the university, but we have tried to glance over a few papers from the field, which have been explained and reviewed done below in the section for Literature Review. We also plan to look into if any other processes that we have learnt can be applied directly into this project or not, trying to first work on our learnings before directly jumping into the coding aspect of the project, trying to read and learn from actual professionals of the field, reports from actual Oncologists and other fellow Researchers from around the world, trying to better understand the terminology and get a better understanding of the topic because it is much more different than we currently learn during our Engineering Courses.

III. LITERATURE REVIEW:-

There is one paper titled ‘Magnetic resonance image-based brain tumour segmentation methods: A systematic review’. This paper doesn’t propose a new algorithm for tumour segmentation. This is a review and comparison of brain tumour segmentation methods in the time period of 2015-2020. It basically covers four major sequences,

- T1-weighted
- T2-weighted
- FLAIR
- T1-Gd

and deals with multimodal MRI. The dataset which we are considering for our project has T1-weighted and T2-weighted along with functional MRIs which have BOLD (Blood Oxygen Level Dependent) information of the brain. T1-weighted is useful for anatomical structure and tumour-related tissue characterization. T2-weighted is useful for visualizing tumour-related abnormalities and tumour core. This paper contains review following segmentation methods,

TABLE I

Method category	Number of studies	Examples
Physics / Mathematics-based	61	Graph cuts, clustering, probabilistic models, active contours, mathematical/biophysical models
Deep Learning-based	156	U-Net, CNN, VGG, DeepMedic, GAN, ResNet, V-Net, etc.
Software / Semi-automatic	6	Software-assisted or semi-automatic segmentation

We are trying to use this paper as a reference to how to apply our own learnings from the course into the designated topic, This is a good starting step because everything has been well defined, but we are still working on understanding some of the wording used throughout the paper, which we are going to look into much deeper from the next week. We have clearly learnt about the few sequences used here but

then we will also try to associate these with the Filtering, Segmentation, Transformation and Restoration techniques in the coming weeks, We believe this serves as a good base for the topic, we will look into further more different papers to conceptually compare how to better apply our concepts too.

Repo Link:-

https://github.com/Rushin-Mod/DIP_2026_P20_G9

REFERENCES

- [1] Bhalodiya, J. M., Lim Choi Keung, S. N., & Arvanitis, T. N. (2022). Magnetic resonance image-based brain tumour segmentation methods: A systematic review. *Digital Health*, 8, 20552076221074122. <https://doi.org/10.1177/20552076221074122>
- [2] “OpenNeuro,” Openneuro.org, 2026. <https://openneuro.org/datasets/ds005003/versions/2.0.0> (accessed Sept. 12, 2026).